Preventive Veterinary Medicine 144 (2017) 112–116

Contents lists available at ScienceDirect

Preventive Veterinary Medicine

journal homepage: www.elsevier.com/locate/prevetmed

Effect of immunologic solutions on sows and gilts on time to stability, and production losses in breeding herds infected with 1-7-4 PRRSv

D.C.L. Linhares[a,*], C. Betlach[b], R.B. Morrison[b]

[a] Iowa State University College of Veterinary Medicine, 2221 Lloyd Veterinary Medicine Center, 1600 16S Street, Ames, IA, 50010, USA
[b] University of Minnesota College of Veterinary Medicine, 385B Animal Science Veterinary Medicine Building, 1988 Fitch Avenue, St. Paul, MN, 55108, USA

ARTICLE INFO

Article history:
Received 1 February 2017
Received in revised form 1 May 2017
Accepted 30 May 2017

Keywords:
PRRSv
Control
Elimination
Whole herd exposure
Live virus exposure
Modified-live virus vaccine
Statistical process control
EWMA
Pigs

ABSTRACT

Porcine reproductive and respiratory syndrome virus (PRRSv) is an economically significant swine pathogen causing production losses in the global swine industry. Clinical impact depends on many factors including the virus itself. One method to sub-type PRRSv is using restriction fragment length polymorphism (RFLP). The RFLP pattern 1-7-4 emerged within the United States swine industry in 2014 and has become prevalent since then. This was a field study that prospectively followed 1-7-4-infected breeding herds (n = 107) and compared time to stability (TTS), time to baseline production (TTBP) and total loss per 1000 sows between herds using modified-live virus vaccine (MLV) on sows and gilts (MLV-MLV), MLV on sows and MLV in addition to field virus exposure on gilts (MLV–MLV/FVE) or not deliberately exposing sows or gilts to PRRSv (Natural-Natural). Analyses were done in SAS 9.4 and results were adjusted by selected co-variates (duration of herd closure, number of previous PRRSv outbreaks of last 3 years, weaning frequency/week, gilt development unit location, herd size and production system). Survival analysis was conducted on TTS and TTBP and regression analysis on total loss. Herds in the Natural–Natural group achieved TTS and TTBP before other herds. Herds in the MLV-MLV/FVE had the longest TTS and TTBP. The total loss was numerically least in MLV–MLV herds (1194 pigs/1000 sows) compared to MLV/MLV-FVE (1810/1000 sows) and Natural–Natural (2671/1000 sows). This study provided additional information to assist veterinarians deciding between methods of exposure to manage PRRSv infection from breeding herds.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Porcine reproductive and respiratory syndrome virus (PRRSv) causes significant economic losses in pig production (Holtkamp et al., 2013; Nieuwenhuis et al., 2012; Neumann et al., 2005). It was estimated that PRRSv reduced reproductive efficiency by 1.44 weaned pigs per breeding female per year, which was equivalent to a cost increment of $ 52.19/sow/year in the entire US swine industry (Holtkamp et al., 2013).

PRRSv continues to expand its genetic diversity (Murtaugh et al., 2010). One way to classify PRRSv is by the restriction fragment length polymorphism (RFLP) typing. The RFLP pattern 1-7-4 emerged within the United States swine industry in 2014 and has become prevalent since then. It has been reported that herds infected with the 1-7-4 PRRSv family have increased production impact when compared to other PRRSv sub-types (Linhares et al., 2015; Alkhamis et al., 2016).

Aiming to reduce PRRSv impact, veterinarians developed several tools to control and/or eliminate virus from pig populations (Corzo et al., 2010). In the North American swine industry it is a common practice to eliminate PRRSv from infected breeding herds without depopulation using methods such as “herd closure” which may be associated with “whole herd exposure” to PRRSv immunologic solutions (Linhares et al., 2014; Torremorell et al., 2003; Torremorell et al., 2002). For the purpose of this study “herd closure” is defined as temporary interruption of replacement pig introduction in the breeding herd, and “whole herd exposure” is deliberately infecting the breeding stock with a replicating PRRSv (the resident virus or an attenuated PRRSv vaccine). The combination of both practices (herd closure and whole herd exposure) is commonly referred as “load-close-expose”.

A previous quasi-experimental study following acutely PRRSV-infected breeding herds undergoing virus elimination by implementing herd closure associated with whole-herd exposure reported a median time to produce PRRS-negative pigs at weaning of 26.6 weeks (range 12–43 weeks) and median time to recover productivity levels (baseline production) of 16.5 weeks (range 0–29 weeks) (Linhares et al., 2014). However, there was considerable

* Corresponding author.
E-mail address: dcl.linhares@gmail.com (D.C.L. Linhares).

http://dx.doi.org/10.1016/j.prevetmed.2017.05.024
0167-5877/© 2017 Elsevier B.V. All rights reserved.

genetic variation of PRRSv isolates in that study and there was no herd *not* implementing herd closure or whole herd exposure.

This study was designed to prospectively follow breeding herds that became infected with 1-7-4 PRRSv and had intent to produce PRRS-negative pigs at weaning after implementing a control/elimination strategy, including use of immunologic solutions on breeding sows and/or incoming gilts. The main variables of interest (study groups) were the PRRS exposure status of the sow farm (MLV vaccination or nothing) and of the gilt pool (MLV vaccine, MLV and LVE, no deliberate exposure). Groups were compared based on the following outcomes: "time to reach stability", "time to recover baseline production" and total loss per 1000 sows.

2. Materials and methods

2.1. Study design

This was an observational study following 107 breeding herds acutely infected with PRRSv belonging to the RFLP pattern 1-7-4 that adopted a strategy to produce PRRSv-negative at weaning (i.e. goal to achieve PRRS stability). The main effect (major factor of interest) was the PRRSv exposure status of sows and incoming gilts. Results were adjusted to other variables including the duration of herd closure (number of consecutive weeks with no gilt introduction following PRRS detection), wean frequency (1, 2, 3+ weaning events per week), location of gilt development unit (on-site or off-site GDU), number of outbreaks in the previous 3 years (0, 1–2, 3+), production system associated with the breeding herd (systems A, B, C, D or E) and herd size (number of inventoried breeding sows). The main effects were compared using the following whole-herd outcomes: time to stability (TTS), time to baseline productivity (TTBP) and total loss. Data was captured using an Excel-based survey, which was sent to the veterinarian assisting the herd.

2.2. Herds and eligibility criteria

The Swine Health Monitoring Project (SHMP) is a network of U.S.A. swine producers that voluntarily agree to share weekly status for different diseases including PRRSv (Perez et al., 2016; Goede et al., 2015). The SHMP was established in 2013 and as of January 2017 had 762 breed-to-wean herds (2,149,963 sows) sharing weekly PRRS status. Veterinarians servicing breed-to-wean herds enrolled in the SHMP in January 2015 were contacted to participate in this study. Enrolled herds fulfilled the following eligibility criteria: (a) sow inventory not expanding or shrinking by more than 10%; (b) diagnostic evidence of PRRS 1-7-4 infection, (c) no diagnostic evidence of infection with an unrelated PRRSv strain during study period, (d) intent to produce PRRSv negative piglets at weaning – in other words, intent to reach stability as defined by (Holtkamp et al., 2011), (e) agreement to sample at least 30 weaning-age pigs for PRRSv RNA by PCR on a monthly frequency starting no later than 12 weeks post detection of PRRS outbreak and (f) willingness to share weekly production performance from the herd. Also, to increase data consistency for the TTBP analysis, enrolled farms had not been infected with porcine epidemic diarrhea virus (PEDv) in the preceding 12 months. Herds were dropped from the study if there was diagnostic evidence of unrelated PRRSv introduction between enrollment and achieving stability (TTS).

2.3. Study groups/key variables of interest

The main variables of interest (study groups) were the PRRS exposure status of the sow farm (MLV vaccination or nothing) and of the gilt pool (MLV vaccine, MLV and LVE, no deliberate exposure):

- Group 1 (MLV – MLV): Sow farm exposure to MLV, and receiving gilts exposed to MLV (n = 41). The sow farm exposure consisted of vaccinating all breeding females.
- Group 2 (MLV – MLV/LVE) Sow farm exposure to MLV, and receiving gilts exposed to MLV and live resident virus (n = 40). The sow farm exposure consisted of vaccinating all breeding females.
- Group 3 (Natural – Natural): No deliberate exposure of PRRSV to sows or gilts (n = 26).

A survey was used to record key demographic information including: herd size (number of breeding sows in the inventory); location of gilt development unit (off site or on site); duration of herd closure in weeks; prior PRRS status (Holtkamp et al., 2011); type of gestation layout (individual stalls or group-housed); frequency of weaning events per week; exposure of breeding sows and/or incoming gilts to PRRSv (resident virus and/or modified-live virus vaccine); on-going exposure of breeding sows and/or incoming gilts to PRRS (resident virus and/or modified-live virus vaccine). Results were adjusted for production system and U.S. geographic region.

2.4. Outcome variables

Time to stability (TTS), time to baseline production (TTBP) and total loss attributed to PRRS infection and subsequent disease control practices were recorded following methods previously established (Linhares et al., 2014). Briefly, herds were assumed to reach stability (i.e. achieve TTS status) when they obtained 4 consecutive negative results on monthly PRRSv RT-PCR tests of 30 piglet blood samples (tested in pools of 1:5). TTS was defined as time from reported PRRS outbreak to the date of the 4th consecutive negative test. TTBP was calculated by the exponentially weighted moving average (EWMA) method, using the 21 weeks prior to PRRS outbreak as baseline period and using kappa of 0.400 and 3 *sigma* to define the control limits. Total loss was an area under the curve calculation where differences from total pigs weaned from the baseline average were summed to represent number of pigs not weaned following PRRS infection, until TTBP was reached.

2.5. Statistical analysis

Survival (Kaplan-Meier for univariate and Cox proportional-hazards regression for multivariate) analysis was used to describe TTS and TTBP metrics. Time zero was defined as the date of reported PRRSv outbreak. Herds were censored if dropped from study between enrollment and reaching the event of interest (TTBP or TTS), or at 52 weeks if not reaching the event by then. General linear model was used to compare total loss per 1000 sows between PRRS control groups. Factors associated with the outcomes (TTS, TTBP or total loss) with $P < 0.200$ in the univariate analysis were entered in the multivariate model and non-significant factors (>0.05) were removed by backward-selection procedure. The treatment factors (main effects) were forced into all the multivariate models. Spearman correlation coefficient was used to compare TTS, TTBP and Total loss. Major model assumptions, including proportional hazards for Cox Regression, normality of residuals for GLM, and linear relationship for numerical predictors were tested and fulfilled. All statistical analyses were performed with SAS 9.4 software (SAS Institute Inc., Cary, North Carolina).

3. Results

Baseline demographic characteristics of the herds enrolled in this study is available on Table 1.

Table 1
Baseline demographic characteristics of the herds enrolled in the study.

Characteristic	Study groups		
	Natural–Natural	MLV–MLV	MLV–MLV/LVE
Number of herds	26	41	40
Mean herd size ± standard deviation	2064 ± 1028	3297 ± 1259	2751 ± 1380
Mean number of weeks from PRRS outbreak detection and gilt introduction ± standard deviation	10.1 ± 7.1 weeks	10.9 ± 6.8 weeks	12.5 ± 8.8 weeks
Wean Frequency	1/week: 22 (85%) 2/week: 2 (8%) 3/week: 2 (8%)	1/week: 14 (34%) 2/week: 8 (20%) 3/week: 19 (46%)	1/week: 33 (83%) 2/week: 7 (18%) 3/week: 0 (0%)
GDU location	On site: 5 (19%) Off site: 21 (81%)	On site: 12 (29%) Off site:29 (71%)	On site: 11 (27%) Off site: 29 (73%)
Number of outbreaks in the previous 3 years	None: 6 (23%) 1 outbreak: 3 (12%) 2 outbreaks: 9 (35%) ≥3 outbreaks: 8 (31%)	None: 11 (27%) 1 outbreak: 6 (15%) 2 outbreaks: 5 (12%) ≥3 outbreaks: 19 (46%)	None: 5 (13%) 1 outbreak: 25 (10%) 2 outbreaks: 18 (45%) ≥3 outbreaks: 7 (18%)

Table 2
Summary of outcomes by study group.

Results	Study groups		
	Natural–Natural	MLV–MLV	MLV–MLV/LVE
Median TTS (25th and 75th percentile)	39.00 (33.00–52.00) weeks	44.00 (33.00–52.00) weeks	47.00 (39.00–52.00) weeks
Median TTBP (25th and 75th percentile)	9.25 (0.00–22.60) weeks	13.00 (7.80–20.10) weeks	20.95 (16.10–23.75) weeks
Mean total loss ± standard deviation	2,671.57 ± 560.69 pigs not weaned per 1000 sows	1,194.66 ± 441.31 pigs not weaned per 1000 sows	1,810.77 ± 529.98 pigs not weaned per 1000 sows

3.1. TTS outcome

Among the 107 herds enrolled, 67 (62.62%) reached stability. There was no difference between study groups on percentage of herds reaching TTS by the end of the observational period (P 0.9244). The overall TTS ranged from 19 to 52 weeks. The median TTS was 44 weeks (25th and 75th percentile 35 and 52 weeks), which is equivalent to 32 weeks (25th and 75th percentile 23 and 40 weeks) when assessed from date of first negative test followed by three consecutive negative tests.

On the univariate analysis, treatment group 3 (natural-natural) had numerically shorter 25th, 50th (median) and 75th percentiles for TTS compared to other groups (P = 0.2323). More specifically, the TTS (median and interquartile range) was 44.00 (33.00–52.00), 47.00 (39.00–52.00), and 39.00 (33.00–52.00) for the MLV-MLV, MLV-MLV/FVE and Natural-Natural, respectively (Table 2). Furthermore, weaning frequency, number of prior PRRS outbreaks and production system were significant (Log rank P = 0.0161 and 0.0202 and 0.0155 respectively). More specifically, weaning 1 or 2 times per week was associated with shorter TTS than weaning 3 or more time per week. Reporting more than 3 prior PRRS outbreaks was associated with longer TTS than 0, 1 or 2 outbreaks. Herds belonging to production system D had a significantly shorter TTS while herds within system B had significantly longer TTS than other systems (systems A, C and E). There was a trend for significance for the "Gilt development unit location", where herds receiving gilts from an on-site GDU had shorter TTS than those receiving gilts from off-site GDU (Log rank P = 0.1631). The range of herd sizes were 714–6,077 inventoried sows. When categorized based on quartiles (14 herds below 1779 sows; 26 herds between 1780 and 2364 sows; 29 herds between 2365 and 4039 sows and 38 herds with more than 4040 sows). Herd size was not significant in the Kaplan-Meier (univariate) analysis (Log rank P = 0.4508).

On the multivariate analysis (Cox proportional hazards regression), the final model included the treatment group (P 0.0175) and production system (P 0.0074). In other words, adjusting by production system, there was a difference on the hazard ratio to achieve TTS between the groups: herds in the MLV–MLV group and MLV-MLV/LVE group had a hazard ratio of 0.603 and 0.192 to achieve TTS compared to herds in the Natural–natural group.

3.2. TTBP outcome

Of the 107 herds, one herd was censored at 28 weeks before reaching TTBP. All the remaining 106 herds (93%) reached TTBP by 36 weeks. The median time and interquartile range (25th and 75th percentiles) were 17.80 and 8.10–22.40 weeks, respectively. There was a significant difference of TTBP between the study groups (Log rank P 0.0121). Herds in the MLV–MLV/FVE had a delayed TTBP (median and interquartile range of 20.95, 16.10–23.75 weeks) compared to herds in the MLV–MLV or Natural–Natural groups (median and interquartile ranges of 13.00, 7.80–20.10 and 9.25, 0.00–22.60 respectively) (Table 2).

The univariate analysis of co-variates revealed that number of prior PRRS outbreaks (Log rank P 0.0002), weaning frequency (Log rank P < 0.0001) and production system (Log rank P < 0.0001) were significant. Moreover, herd size categorized in quartiles had a trend for significance (Log rank 0.1184). TTBP was shorter as number of prior PRRS outbreaks increased, when herds were managed by production systems D or B, and when weaning frequency was ≥3/week. Despite being significant, the effect of herd size on TTBP was not consistent as it was shorter for larger herds (above 4040), followed by herds between 1780 and 2364 sows, then by herds under 1780 sows and finally by herds between 2365 and 4039 sows. GDU location (in site or off-site) was not significant in the TTBP analysis (Log rank P = 0.8588).

The final TTBP multivariate model included weaning frequency (P 0.0050), production system (P 0.0038) and treatment group (P 0.0274). Herds associated with production system D had a hazard ratio of 4.807 (95% confidence limits of 0.866–26.703) to reach TTBP in the following week when compared to herds associated with production system E. Conversely, herds A, B and C had hazard ratio (95% confidence limits) of 0.197 (0.015–2.505), 0.037 (0.003–0.510) and 0.604 (0.272–1.340) respectively when compared to herds on system E. Moreover, herds with weaning frequency of 1 and 2 days had a hazard ratio of reaching TTBP of 0.020 (95% CL 0.002–0.246) and 0.014 (95% CL 0.001–0.184) respectively when compared to herds

weaning at least 3 days per week. Adjusted by weaning frequency and by production system, herds of the MLV-MLV/FVE group and MLV–MLV had a hazard ratio to achieve TTBP of 0.455 (95% CL 0.086–2.415) and 0.215 (95% CL 0.0.045–1.019) respectively when compared to herds in the Natural–Natural group.

TTBP was poorly correlated with TTS (correlation coefficient 0.05626, P 0.5649).

3.3. Total loss outcome

The "total loss" was measured by the number of pigs not weaned per 1000 sows between the time of PRRS detection and herd reaching baseline production (TTBP). Total loss ranged from 0.00 to 5,571.50. The mean and standard deviation was 1,923.02 ± 1,553.87. Adjusted by significant covariates (system and number of prior PRRS outbreaks), total loss trended for significant difference among the study groups (P 0.0969). The mean and standard deviations of total loss for MLV-MLV/FVE, MLV-MLV and Natural-natural groups was 1,810.77 ± 529.98, 1,194.66 ± 441.31 and 2,671.57 ± 560.69 respectively (Table 2).

Total loss had a poor correlation with TTS (0.0155; P 0.8745) and was highly correlated with TTBP (0.8994; P < 0.0001).

4. Discussion

The PRRSv RFLP 1-7-4 family emerged in the U.S. swine industry in 2004 and has been described as highly virulent compared to other PRRSv strains (Alkhamis et al., 2016; Linhares et al., 2015). This was an observational study that characterized the herd-level outcomes of 1-7-4 PRRSv infection on U.S. breeding herds adopting health interventions on sows and/or incoming gilts. The metrics used to report the outcomes included time to stability, probability to reach stability, time to baseline production and total loss/1000 sows. Altogether these metrics provide applied information to veterinarians and producers to understand what to expect and make informed decisions on health interventions of breeding herds that become infected with virulent PRRSv strains.

This study included breeding herds that implemented MLV vaccination (n = 81) and breeding herds that did not deliberately expose sows to PRRSv (Natural–Natural group, n = 26). From breeding herds that implemented MLV vaccination on sows, some herds received gilts previously exposed to MLV only (MLV-MLV group, n = 41) and other herds received gilts previously exposed to both MLV and live field virus (MLV-MLV/FVE group, n = 40). From the Natural–Natural group, veterinarians relied on "natural spread" of PRRSv (as opposed to deliberate exposure/infection), in large part due to gilts coming from PRRS-infected sources.

A previous study with (n = 61) U.S. breeding herds acutely infected with PRRSv attempting to eliminate the virus reported a shorter TTS (26.6 weeks) and higher success rate of achieving stability (80%) (Linhares et al., 2014). Key differences between that population of herds and the one herein reported include that in the previous study (a) all herds implemented herd closure program, (b) there was a greater diversity of PRRSv strains as there were multiple RFLP patterns, and (c) the primary goal of veterinarians was to eliminate PRRSv from the breeding herd, as opposed to control/stabilize the infection. Some of these characteristics may explain the differences in the observed outcomes compared to this study.

Full herd closure, defined as interruption of gilt introduction into the breeding herd until the herd consistently produces PRRS-negative piglets at weaning (Torremorell et al., 2003), was not implemented on herds enrolled in this study. For the herds of this study, the number of weeks between the detection of PRRSv outbreak and introduction on replacement gilts into the breeding herd was captured and recorded as (partial) herd closure. However, the duration of partial closure was not associated with any outcome of interest. One of the objectives of the full herd closure program is to interrupt PRRSv transmission between gilts and sows. By introducing gilts into the breeding herd only when PRRS-protective immunity has been developed in most pigs (i.e. herd immunity), the likelihood of within herd transmission is expected to be less. Field studies are needed to further investigate the role of herd closure on probability of reaching stability and on TTS (i.e. comparing TTS between herds implementing herd closure to those not implementing herd closure, while using the same immunization methods).

TTS and TTBP differed significantly among the 5 production systems. This finding is in agreement with our previous study (Linhares et al., 2014), and supports the hypothesis that specific standard operating procedures (SOPs), along with pig flow or barn layout associated with specific production systems may play a significant role on success rate of disease elimination programs. Studies are needed to assess the effect of specific procedures such as management practices (Lowe et al., 2012) on TTS and TTBP.

This study revealed that the number of weaning events per week was significantly associated with TTBP (significant in the final multivariate model) and was associated with TTS (significant in the univariate model). Herds with ≥3 weaning events per week had a shorter TTBP compared to herds ≤2 weaning events per week. We interpret that by increasing the frequency of weaning events results on loading-out PRRS-affected piglets more frequently. Advantages of more frequent weaning events for the production stand point include the increased consistency of piglet quality (less variation of weight at weaning) and less variation of piglet age at weaning, which facilitates diet and environmental temperature management at the nursery (Main et al., 2004). Disadvantages of increasing weaning events include increased labor and increased herd vulnerability to becoming infected with pathogens, as each weaning event includes contact with trailers, drivers, and other potential biosecurity breaches. Thus, biosecurity procedures should be reinforced at each weaning event and pathogen introduction risk should be evaluated before increasing the frequency of weaning events.

It has been reported that breeding herds with no history of PRRS infection within prior 3 years had increased TTS and TTBP compared to herds with recent infection before detecting PRRSV outbreak. We expected to find similar result in this study. However, after adjusting to other co-variates, number of prior PRRS outbreaks was not significant in the TTS or TTBP analysis.

Gilt development unit (GDU) location (in-site or off-site) was also significant in the univariate TTS analysis but not on TTBP univariate or any multivariate analysis. We expected that herds with in-site GDU would outperform herds with off-site GDU (better TTS, TTBP, total loss), as in-site GDU allows exposure of gilts to similar PRRSv circulating in the recipient breeding herd. This study did not include information of genetic similarity (i.e. ORF-5 or whole genome sequence) of PRRSv used for gilt exposure in comparison to the virus detected in the breeding herd outbreak. Our study detected numerically better breeding herd outcomes (TTS, TTBP, Total loss) when gilts were exposed to MLV alone, compared to when gilts were exposed to both live resident virus and MLV vaccine. We hypothesize that exposing gilts to more virulent PRRSv would affect gilt growing performance, which has been demonstrated to impact lifetime reproductive performance (Williams and Dillard, 2015; Williams et al., 2005; Bortolozzo et al., 2009; Kummer et al., 2009). Further studies are needed to investigate the effect of gilt acclimation to PRRSv on breeding herd immunity to PRRSV and performance.

RFLP patterns are widely used by US veterinarians to describe genetic variability of PRRS viruses. The genetic variability of 1-7-4 PRRSv detected in the US between 2014 and 2015 have

been described (Alkhamis et al., 2016). Phylogenetic comparison between PRRSv-ORF 5 nucleotide sequences of herds enrolled in this study revealed a nucleotide similarity of 98% to 100% (data not shown). Further characterization of genetic diversity among viruses infecting herds of this study requires whole PRRSv genome sequencing. However, whole genome information is three times more costly than RFLP pattern prediction (by ORF 5 sequence) and, to our knowledge, there is no validated method to group viruses based on whole genome information for epidemiological analysis.

Numerically, herds belonging to the Natural–Natural group had the lowest TTS and TTBP. Even though not statistically significant, these herds also had a numerically smaller herd size, which is a characteristic shown to be negatively associated with disease impact and vulnerability to disease introduction. Likewise, herds from the natural–natural group also had a lower frequency of weaning per week (85% of herds had only 1 weaning/week) compared to herds of other groups. Future studies are needed to investigate the role of these variables on TTS, TTBP and total loss.

The median TTS and TTBP times and the interquartile ranges may have been underestimated since there was a large amount of censored observations at the end of the observational period (52 weeks). However, because of censored information, survival analysis was the most appropriate statistical test to measure and compare TTS and TTBP within and between study groups. Porcine epidemic diarrhea virus (PEDV) was endemic in the U. S. swine industry during the research period and PEDV infection results on a significant impact on number of pigs weaned per week. Therefore, we adjusted for this potential confounder by not including herds with PEDv infection in the 12 months prior to the enrollment. However, we did not adjust for other infectious agents with potential effect on the outcomes herein reported.

5. Conclusions

TTS and TTBP were shorter in breeding herds that had natural exposure of gilts and sows to PRRS virus. Herds belonging to the Natural–Natural group had a numerically smaller herd size, mean number of weeks from PRRS outbreak detection and gilt introduction, and weaning frequency, when compared to herds of other groups. Total loss (number of pigs not weaned attributed to PRRS) was numerically less for herds using MLV on sows and gilts following PRRSv infection, which was in agreement with literature, indicating that attenuated PRRSv has a lower clinical/production impact in pig populations as compared to wild type PRRSv. TTS and TTBP were longer and the success rate for reaching stability was lower compared to our previous study. Key differences between a previous study and this study include genetic differences among PRRSv and adoption of herd closure (not implemented on herds of this study). Further studies are needed to elucidate the role of herd closure on herd-level metrics of PRRS control including TTS, TTBP and total loss. Field studies are also needed to better understand how to replicate low TTS, TTBP and total loss of breeding herds acutely infected with PRRSv.

Acknowledgement

Boehringer Ingelheim Vetmedica Inc provided to support the study.

References

Alkhamis, M.A., Perez, A.M., Murtaugh, M.P., Wang, X., Morrison, R.B., 2016. Applications of bayesian phylodynamic methods in a recent U.S. porcine reproductive and respiratory syndrome virus outbreak. Front. Microbiol. 7, 67.

Bortolozzo, F.P., Bernardi, M.L., Kummer, R., Wentz, I., 2009. Growth, body state and breeding performance in gilts and primiparous sows. Soc. Reprod. Fertil. Suppl. 66, 281–291.

Corzo, C.A., Mondaca, E., Wayne, S., Torremorell, M., Dee, S., Davies, P., Morrison, R.B., 2010. Control and elimination of porcine reproductive and respiratory syndrome virus. Virus Res. 154 (1–2), 185–192.

Goede, D., Murtaugh, M.P., Nerem, J., Yeske, P., Rossow, K., Morrison, R., 2015. Previous infection of sows with a mild strain of porcine epidemic diarrhea virus confers protection against infection with a severe strain. Vet. Microbiol. 176 (1–2), 161–164.

Holtkamp, D., Polson, D., Torremorell, M., Morrison, R., Classen, D., Becton, L., Henry, S., Rodibaugh, M.T., Rowland, R.R., Snelson, H., Straw, B., Yeske, P., Zimmerman, J., 2011. Terminology for classifying swine herds by porcine reproductive and respiratory syndrome virus status. J. Swine Health Prod. 19 (1), 44–56.

Holtkamp, D., Kliebenstein, J., Neumann, D., 2013. Assessment of the economic impact of porcine reproductive and respiratory syndrome on United States pork producers. J. Swine Health Prod. 21 (2), 72–84.

Kummer, R., Bernardi, M.L., Schenkel, A.C., Amaral Filha, W.S., Wentz, I., Bortolozzo, F.P., 2009. Reproductive performance of gilts with similar age but with different growth rate at the onset of puberty stimulation. Reprod. Domest. Anim. 44 (2), 255–259.

Linhares, D.C., Cano, J.P., Torremorell, M., Morrison, R.B., 2014. Comparison of time to PRRSv-stability and production losses between two exposure programs to control PRRSv in sow herds. Prev. Vet. Med. 116 (1–2), 111–119.

Linhares, D., Levis, I., Wedel, K., Betlach, C., Morrison, R., 2015. Assessing impact of PRRSv 1–7-4 on wean to finish pig health. In: Allen D Leman Swine Conference, Saint Paul, MN.

Lowe, J.F., DeBuse, N., Morrison, R., Torremorell, M., Yeske, P., 2012. A systematic management strategy for breedingherds based on PRRS herd status. In: AASV Annual Meeting, Denver, CO, pp. 517–522.

Main, R.G., Dritz, S.S., Tokach, M.D., Goodband, R.D., Nelssen, J.L., 2004. Increasing weaning age improves pig performance in a multisite production system. J. Anim. Sci. 82 (5), 1499–1507.

Murtaugh, M.P., Stadejek, T., Abrahante, J.E., Lam, T.T., Leung, F.C., 2010. The ever-expanding diversity of porcine reproductive and respiratory syndrome virus. Virus Res. 154 (1–2), 18–30.

Neumann, E.J., Kliebenstein, J.B., Johnson, C.D., Mabry, J.W., Bush, E.J., Seitzinger, A.H., Green, A.L., Zimmerman, J.J., 2005. Assessment of the economic impact of porcine reproductive and respiratory syndrome on swine production in the United States. J. Am. Vet. Med. Assoc. 227 (3), 385–392.

Nieuwenhuis, N., Duinhof, T.F., van Nes, A., 2012. Economic analysis of outbreaks of porcine reproductive and respiratory syndrome virus in nine sow herds. Vet. Rec. 170 (9), 225.

Perez, A.M., Alba, A., Goede, D., McCluskey, B., Morrison, R., 2016. Monitoring the spread of swine enteric Coronavirus diseases in the United States in the absence of a regulatory framework. Front. Vet. Sci. 3, 18.

Torremorell, M., Moore, C., Christianson, W., 2002. Establishment of a herd negative for porcine reproductive and respiratory syndrome virus (PRRSV) from PRRSV-positive sources. J. Swine Health Prod. 10 (4), 153–160.

Torremorell, M., Henry, S., Christianson, W.T., 2003. Eradication using herd closure. In: PRRS Compendium, Des Moines, IA : National Pork Board, pp. 157–162.

Williams, N., Dillard, D., 2015. How health impacts gilt development and performance. In: Allen D Leman Swine Conference, St Paul, MN.

Williams, N., Patterson, J., Foxcroft, G., 2005. Non-Negotiables of gilt development. Adv. Pork Prod., 281–289.